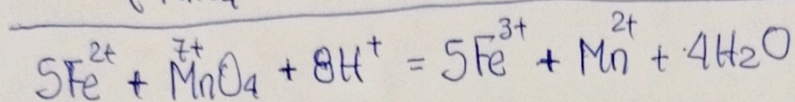
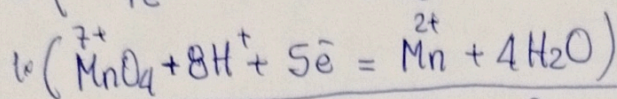
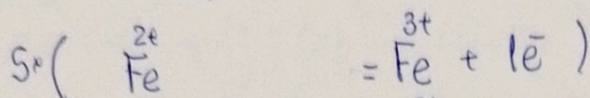
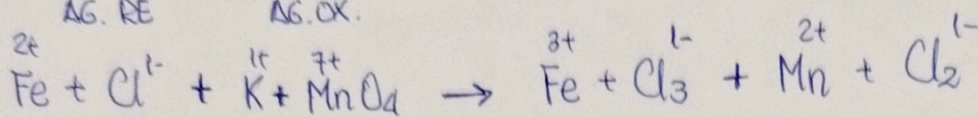
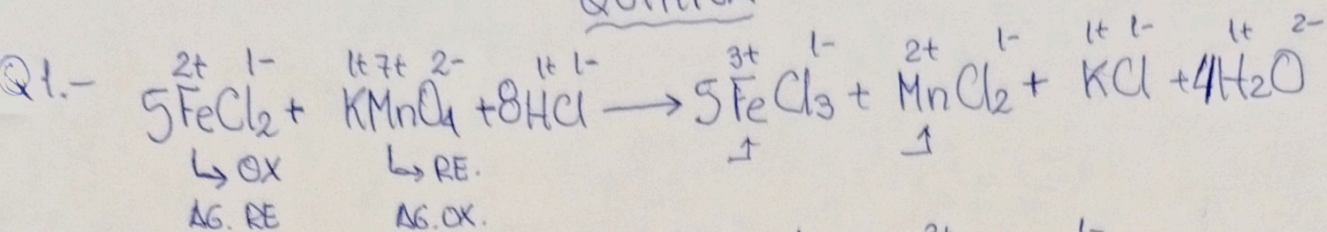
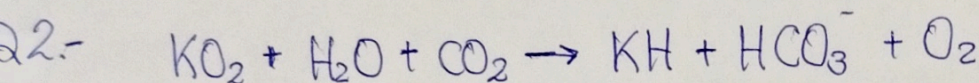


QUÍMICA



$\text{K}: 1=1$
 $\text{Fe}: 5=5$
 $\text{Mn}: 1=1$
 $\text{Cl}: 18=18$
 $\text{H}: 8=8$
 $\text{O}: 4=4$

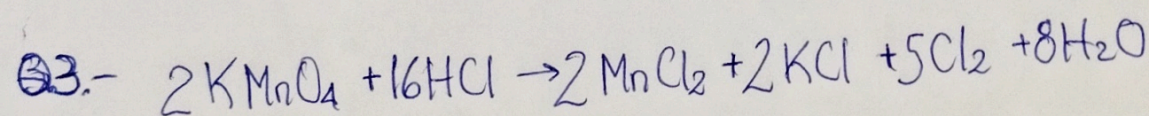
$$X = \frac{\text{Sust. oxidada} - \text{Sust. reducida}}{\text{Agente reductor}} \Rightarrow X = \frac{5-1}{5} = \frac{4}{5} //$$
 Resp. B)



$$100\text{g KO}_2 \times \frac{1\text{mol KO}_2}{71\text{g KO}_2} \times \frac{1\text{mol CO}_2}{1\text{mol KO}_2} \times \frac{44\text{g CO}_2}{1\text{mol CO}_2} \times \frac{1\text{Kg CO}_2}{1000\text{g CO}_2} \times \frac{1\text{día}}{0,95\text{Kg CO}_2} \times \frac{24\text{h}}{1\text{día}} \times \frac{60\text{min}}{1\text{h}}$$

$$= 93,9\text{min} \approx 94\text{min}$$

Resp. A)



$$316,08\text{g KMnO}_4 \times \frac{1\text{mol KMnO}_4}{158\text{g KMnO}_4} \times \frac{16\text{mol HCl}}{2\text{mol KMnO}_4} \times \frac{36,5\text{g HCl}}{1\text{mol HCl}} \times \frac{100\text{g HCl}}{36\text{g HCl}} \times \frac{1\text{cm}^3\text{HCl}}{1,2\text{g HCl}}$$

$$= 1352,19\text{cm}^3\text{HCl}$$

Resp. B)

Q4.-

① DATOS

$$V_1 = 71 \text{ ml}$$

$$T_1 = 0^\circ\text{C} \rightarrow 273,15 \text{ K}$$

$$P_1 = 1 \text{ atm} \rightarrow 760 \text{ TORR}$$

$$T_2 = 27^\circ\text{C} \rightarrow 300,15 \text{ K}$$

$$P_T = 945,13 \text{ TORR}$$

$$P_{H_2O}^\circ = 26,7 \text{ TORR}$$

$$V_2 = ? \text{ ml}$$

$$P_T = P_2 + P_{H_2O}^\circ$$

$$P_2 = 945,13 - 26,7 = 918,43 [\text{TORR}]$$

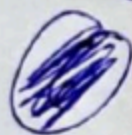
LEY COMBINADA DE LOS GASES.

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$V_2 = \frac{P_1 V_1}{T_1} \cdot \frac{T_2}{P_2} = \frac{760 \text{ TORR} \cdot 71 \text{ ml} \cdot 300,15 \text{ K}}{273,15 \text{ K} \cdot 918,43 \text{ TORR}}$$

$$V_2 = 64,56 [\text{ml}] \rightarrow \textcircled{A}$$

Q5.-



DATOS

$$m_{\text{BaCl}_2 \cdot 2\text{H}_2\text{O}} = ? [\text{g}]$$

$$\overline{M}_{\text{BaCl}_2} = 208,33 \text{ g/mol}$$

$$\overline{M}_{\text{BaCl}_2 \cdot 2\text{H}_2\text{O}} = 244,35 \text{ g/mol}$$

$$m_{\text{BaCl}_2} = 50 \text{ g}$$

$$\% = 12$$

$$50 \text{ g BaCl}_2 \times \frac{12 \text{ g BaCl}_2}{100 \text{ g BaCl}_2} \times \frac{1 \text{ mol BaCl}_2}{208,33 \text{ g BaCl}_2} \times \frac{1 \text{ mol BaCl}_2 \cdot 2\text{H}_2\text{O}}{1 \text{ mol BaCl}_2} \times \frac{244,35 \text{ g BaCl}_2 \cdot 2\text{H}_2\text{O}}{1 \text{ mol BaCl}_2 \cdot 2\text{H}_2\text{O}} =$$

$$m_{\text{BaCl}_2 \cdot 2\text{H}_2\text{O}} = 7,037 [\text{g}] \rightarrow \textcircled{C}$$